ArtBench-10 Generative Modeling: A Comparative Study of VAE, GAN, and Diffusion Models

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Abstract. Nothing yet.

1 Introduction

2 Problem Formulation

3 Methodological Framework

3.1 Data Processing Pipeline

VAE

DCGAN

StyleGAN

Diffusion Models

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Variational Autoencoders (VAE)

Generative Adversarial Networks (GAN)

Diffusion Models

4 Experimental Design

4.1 Hyperparameter Optimization Strategy

We performed hyperparameter optimization with Bayesian optimization (Optuna). To keep the search feasible on standard hardware, tuning was conducted on a development subset comprising 20% of the training data. For each model family, candidate configurations were trained on a training split and selected using performance on a held-out validation split. We minimized the following surrogate objectives: (i) *VAE*: validation reconstruction loss; (ii) *GAN*-based models: validation sum of discriminator and generator losses; (iii) *Diffusion models*: validation denoising loss under a fixed Gaussian noise schedule. The hyperparameter search spaces are summarized in Table 1, and the best configurations found on the studies are reported in Table 2. We used validation losses as surrogates during tuning because FID/KID are computationally expensive and high-variance at the per-trial scale: they require generating large sample sets and repeated feature extraction. In contrast, validation losses are available at each epoch and exhibit lower variance which improves sample efficiency under limited computational resources. We note, however, that GAN losses are far from beeing perfect proxies for perceptual sample quality; therefore, we use this objective only for hyperparameter selection during tuning and rely on qualitative inspection for the selected final parameters. Aditionally, there was no tuning on the StyleGan due to its high computational cost, and we used the default hyperparameters from the original implementation.

Table 1. Hyperparameter search spaces used in Bayesian optimization.

Model	**Parameters**	**Values / Range**
VAE	latent_dim, base_channels, batch_size	$\{16, 32, 64\}$; $\{16, 32, 64\}$; $\{32, 64, 128\}$
	lr, β	$[10^{-4}, 3 \times 10^{-3}]$ log-uniform; $[10^{-6}, 10^{-3}]$ log-uniform
DCGAN	latent_dim, feature_maps, batch_size	$\{128, 256, 512\}$; $\{32, 64\}$; $\{32, 64, 128\}$
	lr, β_1, real_label	$[10^{-4}, 5 \times 10^{-4}]$ log-uniform; $[0.3, 0.7]$; $[0.85, 1.0]$
PixelUNet	model_channels, batch_size, num_timesteps	$\{32, 64, 96\}$; $\{16, 32, 64\}$; $\{500, 1000\}$
	lr	$[10^{-4}, 5 \times 10^{-4}]$ log-uniform
LatentDenoiser	model_channels, num_res_blocks, batch_size	$\{32, 64, 96\}$; $\{2, 3, 4\}$; $\{16, 32, 64\}$
	lr, num_timesteps	$[10^{-4}, 5 \times 10^{-4}]$ log-uniform; $\{500, 1000\}$

Table 2. Best configurations obtained.

Parameter	VAE	DCGAN
Best objective value	0.02658	1.68158
Learning rate	0.0005317	0.0001261
Batch size	32	32
Latent dimension	32	256
Base channels / feature maps	64	64
β (VAE)	1.14×10^{-6}	–
β_1 (Adam, DCGAN)	–	0.4813
real_label (DCGAN)	–	0.9836

Across the selected configurations (Table 2), both the VAE and DCGAN favored a relatively small batch size (32), suggesting that larger batches did not provide a validation advantage under our training budget. For the VAE, the best trial used a low-dimensional latent (32) together with higher base channel capacity (64) and a very small KL weight ($\beta \approx 10^{-6}$), indicating that stronger regularization was detrimental to reconstruction on this dataset. For DCGAN, the best trial used $z = 256$, feature maps 64, a comparatively low learning rate (1.26×10^{-4}), moderate Adam β_1 (0.48), and mild one-sided label smoothing (real_label $\approx$ 0.98), consistent with stabilizing discriminator training.

4.2 Evaluation Protocol

After all sweeps are completed, we use the hyperparameters from table 2 to train the model on the full training set. We compute the FID and KID (the latter computed on 50 subsets of size 100, as required), comparing 5000 generated samples to 5000 real samples from the test set evaluating across 10 different random seeds to account for variability in generation and statistical robustness.

5 Results and Interpretation

5.1 Ablation Studies and Discussion

6 Conclusion

Use of External Resources and AI Tools

This project utilized ArtBench-10 (Kaggle). Gemini assisted in LaTeX formatting and analysis verification.

References

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